

CSCI 5722

Computer Vision

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (64.7 / 100)	47.4 / 100

47	87	+58
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Explores algorithms that can extract information about the world from images or sequences of images. Topics covered include: imaging models and camera calibration, early vision (filters, edges, texture, stereo, optical flow), mid-level vision (segmentation, tracking), vision-based control and object recognition.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	64.7	87.3	+22.6	MEDIUM
Applied Scientist	46.0	92.5	+46.6	HIGH
MLOps Engineer	45.6	82.9	+37.3	HIGH
Machine Learning Engineer	44.4	82.5	+38.1	HIGH
ML Engineer	43.9	77.7	+33.9	HIGH
Data Scientist	39.5	97.6	+58.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 1 of 6 roles

insights

Deep Learning 1 of 6 roles

cnn

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: cnn.

4 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

AI Engineer

64.7 → 87.3 (+22.6 pts)

perception data learning build systems problems technology detection

Applied Scientist

46.0 → 92.5 (+46.6 pts)

computational research literature diffusion interdisciplinary problems data learning

MLOps Engineer

45.6 → 82.9 (+37.3 pts)

detection data text systems learning machine preprocessing media

Machine Learning Engineer

44.4 → 82.5 (+38.1 pts)

data web sets learning analytics tools exploratory design

ML Engineer

43.9 → 77.7 (+33.9 pts)

cnn media data learning machine systems build problems

Data Scientist

39.5 → 97.6 (+58.1 pts)

data sets analytics simply insights social build problems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.